

Summary



**Sébastien
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10/07/1996
French

Data scientist with a PhD in physics and postdoctoral experience in large-scale data analysis, modeling, and uncertainty quantification. Skilled in building scalable data pipelines (Python, SQL, PySpark), applying machine learning and NLP techniques to structured and unstructured data, and delivering reproducible workflows for complex datasets. Experienced in collaborative, international projects and effective at translating technical analyses into actionable, data-driven insights.

Languages: English (fluent), French (native), Italian (intermediate), Spanish (intermediate)

Modelling & Analytics: Physical and statistical modelling, uncertainty quantification, sensitivity and scenario analysis, model validation.

Programming & Data: Python, SQL, Fortran, C/C++, R, Matlab.

Data Science, ML & NLP: pandas, NumPy, scikit-learn, PyTorch, PySpark, applied NLP (text preprocessing, embeddings, semantic search, retrieval), data visualization (Matplotlib, seaborn), dashboards (Streamlit, Power BI).

Engineering & Infrastructure: Docker, APIs, cloud fundamentals (AWS), HPC workflows (Slurm), parallel computing (OpenMP, OpenACC, CUDA).

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Work Experience

Oct 2022 – Oct 2025 **Data Scientist (PostDoctoral Researcher)** Université Libre de Bruxelles, BELGIUM

Sept 2022 – Oct 2022 **PostDoctoral Researcher – Physics** University of Geneva, SWITZERLAND

Sept 2018 – Aug 2022 **Doctoral Researcher – Physics** University of Geneva, SWITZERLAND

Data Science Projects

Statistical Modeling & Scalable Data Pipelines [Python](#), [Fortran](#), [HPC](#), [Machine Learning](#)

- Built a scalable pipeline to run and analyze large-scale simulations (10k+ runs) using legacy Fortran code wrapped in Python; automated deployment on HPC clusters (Lucia Tier-1 HPC) using Slurm batching system.
- Developed and applied a statistical framework to quantify and propagate uncertainties, enabling robust predictions from high-dimensional data. Framework now used by international teams.
- Delivered reproducible workflows and end-to-end tooling to extract actionable insights from complex simulations.
- Collaborated with multidisciplinary teams to translate technical results into usable observables across domains.

Enhancing Legacy Software with Modern Data Science [Python](#), [Fortran](#), [Data Viz](#), [CLI Tools](#), [Git](#)

- Led the design and integration of new physics modules into a legacy simulation code, improving performance and expanding capabilities for broader use cases.
- Modernized data analysis interface with Python and dashboards (NumPy, pandas, seaborn, Power BI, Streamlit), enabling rapid visual exploration.
- Developed robust CLI tools and notebooks for data wrangling and visualization, enhancing reproducibility and team collaboration.
- Demonstrated autonomy and leadership in driving complex projects from concept to publication.

NLP & Retrieval-Augmented Generation (RAG) Systems ([Link to Interactive demo](#)) [Python](#), [NLP](#), [Transformers](#), [Embeddings](#), [Vector Search](#), [LLMs](#), [Streamlit](#)

- Built an end-to-end NLP pipeline for semantic search over large document collections, including text preprocessing, embedding generation, and similarity-based retrieval.
- Designed and implemented a Retrieval-Augmented Generation (RAG) workflow combining vector search with large language models for contextual question answering.
- Evaluated and compared classical NLP, embedding-based retrieval, and LLM-augmented approaches, analysing trade-offs in accuracy, robustness, and latency.
- Delivered interactive prototypes and dashboards (Streamlit) to expose search and Q&A functionalities, enabling rapid iteration and user-driven evaluation.

Education

2022
PhD in Physics
University of Geneva, Switzerland

2018
Master in Physics
Université Grenoble-Alpes, France

2016
Bachelor in Fundamental Physics
Université Pierre et Marie Curie, Paris VI, France

[Full list of publications](#) / [Full list of conferences](#)
(22 papers incl. 9 first author) / (15 talks in 10 different countries)